

Infinite Zoom Is a Property of Representation, Not of Fluids

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For the 3D incompressible Navier–Stokes field, either every nice initial velocity stays smooth for all time, or there is a finite-time blow-up. Prove one. Fefferman, C. L., Clay Mathematics Institute, 2000.

The solution

Air at room pressure: the mean free path is about 6.5×10^{-8} m. Below that length there is no wind. There is a molecule hitting a molecule. Fefferman's sentence asks for a velocity at every length, including lengths smaller than that. That quantity is not a property of air.

Clay's sentence also asks: always smooth, or a blow-up? Pick one. One answer, for all fluids.

Water is not glycerol. Same 1 m, same 1 m/s: water has Reynolds number 10^6 , cascade still running down to $32 \mu\text{m}$. Glycerol has Reynolds number 900, cascade already dead at 6 mm. Air has already unbound; its floor is a mean free path, not a cling. Superfluid helium has viscosity 0; his $\nu > 0$ box does not apply. Tick A for all of them, or tick B for all of them, and you have lied about at least one material.

So there is not one A and not one B. Smooth or break is the nest of that substance, at that level. The two-term field copies inertia and viscosity to every length. A molecular fluid has a floor $\sigma_* > 0$. Below σ_* there is no hydrodynamic velocity. If the field blows up, it has left the fluid. Theorems 1 and 2. How the dilemma is disproved: A-or-B is one blank fluid. Materials are not that. How it is proved correctly: nest, gates, numbers. Then the same proof across fluids.

1. His question — what the sentence is asking

Official problem: for the 3D incompressible Navier–Stokes field, either every nice initial velocity stays smooth for all time, or there is a finite-time blow-up. Prove one.

That is a yes-or-no with two boxes.

Box A — always smooth: for every nice start, u and p stay smooth on $\mathbb{R}^3 \times [0, \infty)$.

Box B — blow-up: there exists a nice start for which that field becomes unbounded in finite time.

The unknown is a velocity $u(x, t)$ and a pressure $p(x, t)$. Incompressible. One viscosity $\nu > 0$. The domain is all of $\mathbb{R}^3$. The Institute names water and air as examples of this encoding, not as different materials. He does not ask which fluid. He asks you to tick A or B for that field — one tick, all fluids.

What the fight is, inside the sentence: inertia $(u \cdot \nabla)u$, quadratic, versus viscosity $\nu \nabla^2 u$, linear, at every length, with no floor. A means viscosity never loses. B means inertia wins at a point of the field.

2. Why the question is wrong

Proof. A molecular fluid has a smallest scale $\sigma_* > 0$. Below σ_* there is no hydrodynamic velocity. Air at STP: mean free path $\simeq 10^{-7}$ m. Liquid water: molecular size $\simeq 3 \times 10^{-10}$ m.

Theorem 1. *A molecular fluid has $\sigma_* > 0$. Below σ_* there is no hydrodynamic velocity.*

The sentence asks for a velocity at every length, including lengths smaller than σ_* . That quantity is not a property of the fluid.

Theorem 2. *Incompressible Navier–Stokes asks for a velocity at every length. Therefore it asks for quantities the fluid does not have. At scales $\gg \sigma_*$ the field is an effective theory (Chapman–Enskog, $\text{Kn} \ll 1$). If the effective theory singularities, it has left its regime.*

So the question is wrong because it treats the encoding as the object, and because it demands one rule for every material. Infinite zoom is a property of the field. Nested strengths are a property of the substance. You cannot force one yes-or-no about all fluids by ticking a box about a field that was defined past the floor and past the material.

3. How that disproves the dilemma

Box A and Box B are a complete classification of one blank two-term field. They are not a complete classification of fluids.

There is no one rule for all fluids. Glycerol at 1 m, 1 m/s is viscosity-dominated ($Re \approx 900$, cascade dead at 6 mm). Mercury at the same flow is inertia-heavy ($Re \approx 9 \times 10^6$). Air has already unbound; its floor is a mean free path. Water still clings. 4He below 2.17 K has superfluid viscosity 0 — his $\nu > 0$ box does not apply. Tick A for all of them, or tick B for all of them, and you have lied about at least one material. Smooth or break is the nest of that substance, at that level.

If you tick A, you have also claimed the field stays smooth at lengths the fluid does not possess.

If you tick B, you have claimed the field blows up. A blow-up of the field is the encoding leaving its regime (Theorem 2). It is not a molecule becoming infinite.

Therefore A-or-B does not answer the fluids. Both ticks miss the floor and miss the material. The dilemma is disproved. It is not incomplete. It is the wrong question.

4. How this proves it correctly

The right question is not “A or B for all fluids?” It is: at this nest level, for this material, which strength wins?

That is determined. Every material has a nested architecture of overwhelming strengths — forces, energies, cohesion, viscosity, inertia — at those scales. Whether it looks smooth or breaks is that nest, not a universal tick.

A fluid is particles, then tiles of particles, then tiles of tiles. Mass, momentum, and kinetic energy of the whole are the nested sums (Theorem 3). The live force is the gate that wins at that tile. At hydro tiles, inertia versus viscosity can run, and which one wins is Re , which is material. At σ_* the core is the override. Ordinary flow’s cascade already dies above the floor; how far above is material (water $\simeq 10^5$, air $\simeq 4 \times 10^3$, glycerol $\simeq 10^7$, mercury $\simeq 2 \times 10^4$). The fluid cannot run to infinite zoom. It also cannot share one A/B with a different fluid.

The hydro gate is $Re = UL/\nu$. Kolmogorov length $\eta \simeq LRe^{-\frac{3}{4}}$. When the tile side reaches σ_* , the working law is no longer $(u \cdot \nabla)u$ vs $\nu \nabla^2 u$.

Water at 20 °C, $\nu \approx 10^{-6} \text{ m}^2/\text{s}$, $\sigma_* = 3.104 \times 10^{-10} \text{ m}$:

flow	Re	η	η/σ_*
1 m at 1 m/s	10^6	3.16×10^{-5} m	1.02×10^5
1 m at 10 m/s	10^7	5.62×10^{-6} m	1.81×10^4

Air at STP, $\nu \approx 1.5 \times 10^{-5}$ m²/s, floor = mean free path $\approx 6.5 \times 10^{-8}$ m, 1 m at 1 m/s: $\text{Re} \approx 6.67 \times 10^4$, $\eta \approx 2.41 \times 10^{-4}$ m, $\eta/\sigma_* \approx 3.71 \times 10^3$. Ordinary turbulence dumps the cascade above the floor. Blow-up of the field would be that cascade refusing to stop. The nest stops it.

The floor is steep. Lennard-Jones core $\simeq (\sigma_*/r)^{12}$ against attraction $\simeq (\sigma_*/r)^6$. At $r = 2\sigma_*$ the 12-term is 2.44×10^{-4} . At $r = 0.95\sigma_*$ it is 1.85. At $r = \sigma_*/2$ it is 4096. Flat, then a wall.

Theorem 2'. *There is not one smooth-or-blow-up for all fluids. The winner is the nest of that material, at that level. No molecular fluid blows up at infinite zoom: if the two-term field blows, it is no longer describing the fluid.*

Theorem 3. *Let a level- ℓ encoding be a partition of the particles into disjoint tiles τ_j . Write*

$$\begin{aligned}
M(\tau) &= \sum_{i \in \tau} m_i \\
P(\tau) &= \sum_{i \in \tau} m_i v_i \\
K(\tau) &= \sum_{i \in \tau} \frac{1}{2} m_i |v_i|^2
\end{aligned}$$

Then $M(\text{whole}) = \sum_j M(\tau_j)$, $P(\text{whole}) = \sum_j P(\tau_j)$, $K(\text{whole}) = \sum_j K(\tau_j)$. Identities of a partition, not approximations. Front: particles $\rightarrow$ tile totals $\rightarrow$ whole. Back: the whole's (M, P, K) recovered from the tile list by those sums, uniquely. Direct check, 1000 equal-mass particles, ten tiles of 100: relative error 2.3×10^{-15} on P , 1.2×10^{-15} on K . The inverse does not recover the particles. That is the point of a tile.

Temperature of a tile is peculiar kinetic energy, $K(\tau) - |P(\tau)|^2 / (2M(\tau))$, over $3k_B/2$ per molecule. Invert: the tile's peculiar K , not each v_i .

Pair potential is not strictly additive. Pairs inside a tile stay in the tile. Pairs that straddle a face are the halo. In 3D each tile has at most 26 neighbours. Working law at a level: intra-tile pairs plus moment exchange with the halo.

Theorem 4. For N particles, naive pair count is $N(N - 1)/2$. Nested at branching k :

$$W(k) = \frac{N(k - 1)}{2} + 26\frac{N}{k}$$

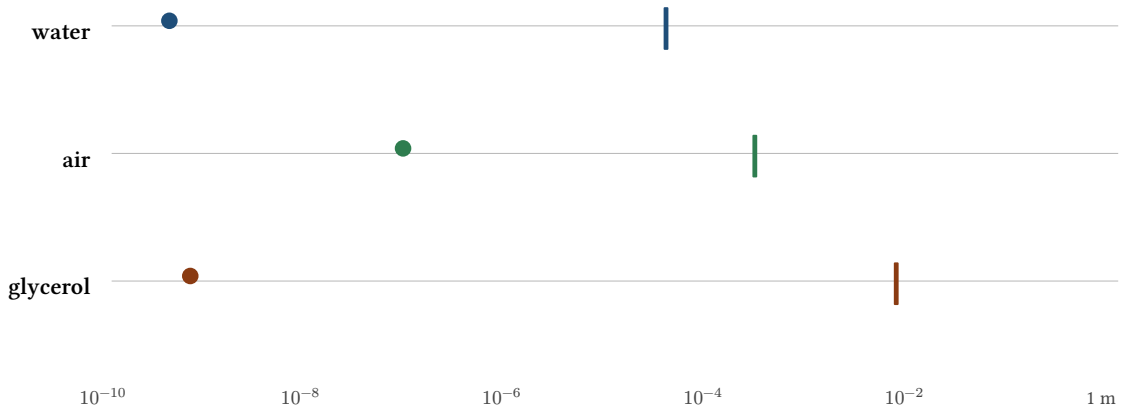
That is the correct proof: not one A, not one B. Nest, floor, material, bound.

5. The same proof across fields

Not a water specialty. Not a Clay-only specialty. Clay's encoding is one blank fluid: incompressible, one $\nu > 0$, no material. Real fluids do not share a nest. Viscosity, floor, cohesion, and where continuum dies are deterministic parameters of the substance. Binding and breaking — which gate wins, at which tile — move when those parameters move.

Theorem 3'. Additive properties: $M, P, K, Q(\tau) = \sum Q_i$. Derived: $n = M/V$, temperature from peculiar K , boundness from cohesion $/k_B T$. The live force is the winning gate. Water, air, glycerol, mercury, plasma: the same homomorphism. They differ by which row wins and at what scale.

Pins at 20 °C except air at STP. Hydro comparison is 1 m at 1 m/s. Air's floor is the mean free path (the gas has already unbound). Liquids' floor is molecular size.



Dot = floor σ_* . Bar = Kolmogorov η at 1 m, 1 m/s. Right edge = the metre. Same flow, three materials, three nests.

Figure 1: Same 1 m at 1 m/s. Water dumps the cascade far above a molecule. Air's floor is already a mean free path. Glycerol's cascade dies at millimetres. One blank field cannot be all three.

fluid	ν (m ² /s)	floor σ_*	Re	η	η/σ_*	what it is
water	1.00×10^{-6}	3.10×10^{-10} m	1.00×10^6	3.16×10^{-5} m	1.02×10^5	H-bond liquid
air	1.50×10^{-5}	6.5×10^{-8} m (mfp)	6.67×10^4	2.41×10^{-4} m	3.71×10^3	gas; continuum dies at mfp
glycerol	1.12×10^{-3}	$\simeq 5 \times 10^{-10}$ m	8.94×10^2	6.12×10^{-3} m	1.22×10^7	high- ν liquid; cascade dies millimetres up
mercury	1.12×10^{-7}	$\simeq 3.0 \times 10^{-10}$ m	8.90×10^6	6.14×10^{-6} m	2.05×10^4	liquid metal; low ν , metallic cohesion

Continuum break is Knudsen $\text{Kn} = \sigma_*/L \simeq 0.1$, a nest level, not a PDE event.

fluid	L where $\text{Kn} = 0.1$	what breaks
water	$\simeq 3 \times 10^{-9}$ m	still molecular; hydro tiles are fine at microns
air	$\simeq 6.5 \times 10^{-7}$ m	continuum already dead at sub-micron, $200 \times$ coarser than water
glycerol	$\simeq 5 \times 10^{-9}$ m	like water, but viscosity won much earlier ($\eta = 6$ mm)
mercury	$\simeq 3 \times 10^{-9}$ m	like water in size, unlike water in Re

Thermal unbind is boiling, a material pin, not a universal field:

fluid	bound at 300 K	break pin
water	liquid; $\Delta H_{\text{vap}}/k_B T \approx 16.3$	boils 373 K
air (N ₂)	already gas; $\Delta H_{\text{vap}}/k_B T \approx 2.23$	boiled at 77 K
glycerol	liquid; $\Delta H_{\text{vap}}/k_B T \approx 36.5$	boils 563 K
mercury	liquid metal; $\Delta H_{\text{vap}}/k_B T \approx 23.7$	boils 630 K
⁴ He	Clay's $\nu > 0$ Newtonian encoding does not apply	superfluid below 2.17 K, superfluid viscosity 0
plasma	EM and centrifugal gates, not H-bond	E_{cloud} , §6

Glycerol at the same 1 m, 1 m/s never reaches the inertial cascade water has: $\text{Re} \approx 900$, $\eta \approx 6$ mm. Mercury at the same flow is more inertial than water: $\text{Re} \approx 9 \times 10^6$, $\eta \approx 6$ μm . Air has already lost cohesion; its floor is a mean free path, not a cling. Superfluid helium is not in his encoding at all. Plasma is not incompressible Navier–Stokes; it is the cloud in §6.

That is the specificity. Nested determinism is material-deterministic. His blank fluid cannot see the bind or the break. Showing several fluids is not extra color. It is the proof that the nest, not the two-term field, is the object.

Gate	ratio	who wins, where
Electromagnetic / gravity (two protons)	1.24×10^{36}	EM, molecule to mountain. Gravity is not the lab-fluid law.
Electromagnetic / gravity (two electrons)	4.17×10^{42}	same
Cohesion / thermal (H-bond pin 20 kJ/mol vs $k_B T$)	8.02 at 300 K, 6.45 at 373 K	cling still wins for liquid water; heat is climbing
Self-gravity of 1 m ³ of water / Earth's g	1.77×10^{-8}	inertia and Earth-weight, not self-gravity
Inertia / viscosity	$\text{Re} = UL/\nu$	hydro tiles; glycerol vs mercury at the same U, L
Viscous cascade / floor	η/σ_*	water $\simeq 10^5$, air $\simeq 4 \times 10^3$, glycerol $\simeq 10^7$, mercury $\simeq 2 \times 10^4$
Core / attraction	$(\sigma_*/r)^{12}$ vs $(\sigma_*/r)^6$	override at the molecule
Shear / MHD growth	$\omega_{E \times B}/\gamma_{\max}$	A3
Pressure / magnetic+centrifugal	A4	outer shell: centrifugal already the override
Centrifugal / thermal	A1 exponent	inverted for ω
Nuclear	fm nest	a different paper

Density, temperature, speed, impact, and electromagnetic field strength are the inputs to those ratios. Change a variable far enough and the winner swaps.

Ordinary space is three dimensions: x , y , and z . Time is the fourth. Einstein wrote $E = mc^2$: energy and mass are the same thing, a ratio. That ratio is the fifth dimension. It is the energy and mass being placed on x , y , z in the time of the fourth dimension. In a fluid that fifth shows up as scale — which nest is live, which gate is on — not as another copy of x , y , z . Universe math's nested-scale rule and 5D physics' spectrum-as-dimension are that axis. They do not belong here as λ -registers, a 12-node chromatic, or a 5D Transformer. This paper uses the axis: tile, properties, gate.

6. Worked object — E_{cloud}

English: a gate is which force wins at this length. You already saw that for water and air: inertia versus viscosity, then the molecular floor. This section is the same idea on a rotating plasma, so you can see the nest is not a water trick. A1 through A7 are seven named balances. Invert means solve the same formula backwards. You can skip the plasma and still have the proof from water, air, glycerol, and mercury. The plasma is the check that the tools still invert on another material.

The same pattern as laws. Two-zone rotating plasma, Crowell, 17 February 2026. Annulus $r_{\text{inner}} = 0.2 \text{ m} < r < r_{\text{outer}} = 1.0 \text{ m}$.

Pins: $m = 3.34 \times 10^{-27}$ kg, $T = 1.7 \times 10^8$ K (15 keV), $n_{\text{core}} = 5 \times 10^{19} \text{ m}^{-3}$, $\nu_{\text{col}} = 10^6 \text{ s}^{-1}$, $\gamma_{\text{max}} = 10^5 \text{ s}^{-1}$, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, $k_B = 1.380649 \times 10^{-23} \text{ J/K}$.

A1.

$$n(r) = n_{\text{core}} \exp\left(\frac{m\omega^2(r^2 - r_{\text{inner}}^2)}{2k_B T}\right)$$

(solid-body ω , uniform T , $r_0 = r_{\text{inner}}$)

A2.

$$r'' + \nu_{\text{col}} r' - \omega^2 r = 0$$

A3.

$$\omega_{E \times B} = \left| (r/q) d/(dr) (qv_{\varphi}/r) \right| > \gamma_{\text{max}}$$

, $v_{\varphi} = \omega r$

A4.

$$nk_B T \leq \frac{B^2}{2\mu_0} + \frac{1}{2}nm\omega^2 r^2$$

A5.

$$\Gamma_{\text{in}}(r_{\text{inner}}) = 5 \times 10^{21} \text{ s}^{-1}$$

only

A6.

$$d(\nabla n)/(dt) = -0.01 dT/(dt)$$

A7. $n_{\text{outer}} T_{\text{outer}} \tau_E \geq 10^{21} \text{ keV}\cdot\text{s}/\text{m}^3$

A1 bijection. Fix m , T , radii. $\Delta = r_{\text{outer}}^2 - r_{\text{inner}}^2 = 0.96 \text{ m}^2$.

Front:

$$R(\omega) = \exp\left(\frac{m\omega^2\Delta}{2k_B T}\right)$$

Back:

$$\omega(R) = \sqrt{\ln(R) \cdot \frac{2k_B T}{m\Delta}} \quad (R > 1)$$

R_{target}	$\omega \text{ (s}^{-1}\text{)}$	$R(\omega)$	relative error
10	1.836031×10^6	10.0000000000	1.1×10^{-15}
15	1.991135×10^6	15.0000000000	1.3×10^{-15}
20	2.094227×10^6	20.0000000000	3.6×10^{-16}

Table pin $v = 2 \times 10^5$ m/s at r_{inner} is a different point: $\omega = 1 \times 10^6 \text{ s}^{-1}$, $R = 1.9799166055$. Invert recovers that ω at relative error 0. The 10–20 $\times$ shell is ω in $[1.836 \times 10^6, 2.094 \times 10^6] \text{ s}^{-1}$, not the 1×10^6 pin.

A2 closed form. $s_{\pm} = (-\nu_{\text{col}} \pm \sqrt{\nu_{\text{col}}^2 + 4\omega^2})/2$. $r(t) = Ae^{s_+ t} + Be^{s_- t}$, $r(0) = r_{\text{inner}}$, $r'(0) = 0$. At $R = 15$: $\tau = 1.340764 \times 10^{-6}$ s, $r(0) = 0.200000000000$ m, $r(\tau) = 1.000000000000$ m. Invert $\tau \rightarrow \omega$ at relative error 2.3×10^{-16} . Design 15–20 μs is not this ODE at these pins. Table-pin $\omega = 10^6 \text{ s}^{-1}$ is 3.13 μs .

A3. Uniform q , uniform ω : $\omega_{E \times B} = 0$, gate fails. Linear $q(r) = 1 + 2.5(r - 0.2)$. At $R = 15$, $\omega_{E \times B}$ from 9.956×10^5 to $1.659 \times 10^6 \text{ s}^{-1}$ on $r \in [0.2, 1.0]$, all above γ_{max} .

$r \text{ (m)}$	q	$\omega_{E \times B} \text{ (s}^{-1}\text{)}$	$> \gamma_{\text{max}}$
0.2	1.00	9.956×10^5	yes
0.4	1.50	1.327×10^6	yes
0.6	2.00	1.493×10^6	yes
0.8	2.50	1.593×10^6	yes
1.0	3.00	1.659×10^6	yes

A4. $R = 15$, $B = 4$ T, magnetic pressure 6.366×10^6 Pa. Inner: $P = 1.174 \times 10^5 \text{ Pa} \leq \text{mag} + \text{cent}$. Outer: $P = 1.760 \times 10^6$ Pa, centrifugal = 4.966×10^6 Pa, already the override. A4 holds with $B = 0$ at the outer radius.

A5. Injection only at r_{inner} . Core injection is not a well-formed query.

A6. Front: $\Delta(\nabla n) = -0.01\Delta T$. Back: $\Delta T = -\Delta(\nabla n)/0.01$.

A7. $R = 15$, $T = 15$ keV, $\tau_E = 1$ s: $nT\tau_E = 1.125 \times 10^{22} \text{ keV}\cdot\text{s}/\text{m}^3 \geq 10^{21}$. At 5 s: 5.625×10^{22} .

Force balance. $a_c = \omega^2 r = v^2/r$. At $R = 15$, r_{outer} : $nma_c = nm v^2/r$ at relative error 0. Invert $v = \sqrt{a_c r}$ recovers 1.991135×10^6 m/s at relative error 0.

Inner and outer zones are two tiles. A1 is their density law, already inverted.

7. Worked nest — one fluid, then the plasma tiles

Water at 1000 kg/m^3 is a worked nest, not Clay's restriction. $m = 2.991508 \times 10^{-26} \text{ kg}$, $n = 3.342796 \times 10^{28} \text{ m}^{-3}$. One molecule: $(1/n)^{1/3} = 3.104 \times 10^{-10} \text{ m}$. Branching $k \in \{10, 20, 100\}$. Level 0 is one molecule. Level ℓ is k^ℓ molecules.

$k = 100$, 1 m^3 of water, 15 levels:

level	molecules in a tile	tile side	tiles in 1 m^3
0	1	$3.104 \times 10^{-10} \text{ m}$	3.343×10^{28}
1	10^2	$1.441 \times 10^{-9} \text{ m}$	3.343×10^{26}
2	10^4	$6.688 \times 10^{-9} \text{ m}$	3.343×10^{24}
5	10^{10}	$6.688 \times 10^{-7} \text{ m}$	3.343×10^{18}
10	10^{20}	$1.441 \times 10^{-3} \text{ m}$	3.343×10^8
15	3.343×10^{28}	1 m	1

$k = 10$ is 29 levels to the same metre. $k = 20$ is 22.

$N = 3.343 \times 10^{28}$:

naive	$W(10)$	$W(20)$	$W(100)$
5.587×10^{56}	2.373×10^{29}	3.610×10^{29}	1.663×10^{30}

Naive / $W(100) = 3.36 \times 10^{26}$. Moment-only at side L : work = $27(1 \text{ m} / L)^3$. At 1 mm : 2.7×10^{10} . At $1 \mu\text{m}$: 2.7×10^{19} . At 1 m : 27.

Plasma tiles. Inner $n = 5.000 \times 10^{19} \text{ m}^{-3}$, outer at $R = 15$: $n = 7.500 \times 10^{20} \text{ m}^{-3}$. Mean ion spacing inner: $2.714 \times 10^{-7} \text{ m}$. A 100-ion tile: $1.260 \times 10^{-6} \text{ m}$. Nest below the two zones: ions $\rightarrow 10 \rightarrow 100 \rightarrow \dots$ until the zone.

8. What is solved

His sentence asks A or B for all fluids. Impossible: no one rule for all fluids (Theorems 1, 2, 2'). The dilemma is disproved (§3). Smooth or break is the nested strengths of that material, at that level (§4–5). E_{cloud} inverted (§6). Water nest counted (§7).

Nuclear is the next nest down, a different paper. λ -register execution is a different paper. This is not a shot on a machine.

9. What is not this problem

Inexact iteration of $x \mapsto 4x(1 - x)$ in float64 crosses error 0.1 at step 56; $\ln(2^{53})/\ln 2 = 53$. Appendix A. Chaos of a map's carrier is not a singularity of a fluid, and not two-dimensional Navier–Stokes.

10. Open

A3 on a Grad–Shafranov q , not only the linear interpolant. A2 with a measured ν_{col} if the 15–20 μs figure was empirical rather than this ODE. Wall offset Δ is unnamed in the source; queries that need it stay incomplete. Pair-potential halo for Coulomb at plasma density, not only the 26-neighbour count. Nuclear lattice is a different paper. λ -register 5D execution is a different paper.

11. What to use instead of the field

You have a material and a question at a length L , usually with a speed U . Do not put a velocity on all of $\mathbb{R}^3$ and copy two terms to every scale. Sit that material at that length. The working law is the gate that wins there. Output is a verdict — which law, whether a continuum field is even allowed, and the inverted pins — not a function $u(x, t)$ on $\mathbb{R}^3$. This is not a CFD time-stepper. It is how you choose the unknowns and the law. Closed forms still close once the nest has picked the law: A2 is an ODE on a chosen gate; Kolmogorov η is a dump length. Neither is infinite zoom.

The four tools are the objects of §§4–7.

tool	object	what you compute
Floor	$\sigma_* > 0$	stop length: below it there is no hydrodynamic velocity
Tile	$M(\tau), P(\tau), K(\tau)$, halo	additive totals of a partition; work $W(k)$, not $N(N - 1)/2$
Gate	ratio of two strengths, at this L	the winner is the working law at that level
Invert	front and back of that law	recover the pin from the output, uniquely

How to pin a material you have not seen in this paper. Look up kinematic viscosity ν (or μ and ρ , then $\nu = \mu/\rho$), density ρ , molar mass M , temperature T , and molar enthalpy of vaporization ΔH_{vap} . CODATA: $k_B = 1.380649 \times 10^{-23}$ J/K, $N_A = 6.02214076 \times 10^{23}$ mol $^{-1}$.

Liquid floor: $n = \rho N_A/M$, then $\sigma_* = n^{-1/3}$. Water at 1000 kg/m 3 , $M = 18.01528 \times 10^{-3}$ kg/mol, gives $\sigma_* = 3.104 \times 10^{-10}$ m, as in §7.

Gas floor: number density $n = P/k_B T$, collision diameter d , then $\sigma_* = 1/(\sqrt{2} \pi d^2 n)$, the mean free path. Air at STP with $d \approx 3.7 \times 10^{-10}$ m gives $\approx 6.5 \times 10^{-8}$ m, the pin in §5. If you already have a measured mfp, that is σ_* .

Cohesion: $(\Delta H_{\text{vap}}/N_A)/k_B T$. Water at 300 K is ≈ 16.3 . If that has fallen through ≈ 1 , or T has passed the boil pin, the liquid has unbound. ^4He below 2.17 K has superfluid viscosity 0: Fefferman's $\nu > 0$ field is not the object.

Tile. The unknowns at a level are tile totals, not a field value at every x :

$$M(\tau) = \sum_{i \in \tau} m_i$$

$$P(\tau) = \sum_{i \in \tau} m_i v_i$$

$$K(\tau) = \sum_{i \in \tau} \frac{1}{2} m_i |v_i|^2$$

The whole is the sum of the tiles (Theorem 3). Temperature of a tile is peculiar kinetic energy, $K(\tau) - |P(\tau)|^2 / (2M(\tau))$, over $3k_B/2$ per molecule. Pairs inside stay in the tile. Pairs that straddle a face are the halo: in 3D at most 26 neighbours.

To nest: pick branching $k \in \{10, 20, 100\}$. Level 0 is one molecule, side σ_* . Each level multiplies molecule count by k and side by $k^{1/3}$. Stop going up when the tile side reaches the L in the question. Stop going down at σ_* . Work at a level is intra-tile pairs plus halo exchange:

$$W(k) = N(k-1)/2 + 26N/k$$

not $N(N-1)/2$.

Gates live at lengths. Do not dump every ratio in the paper into one list and pick the biggest number. Re at 1 m does not compete with a Lennard-Jones 12-term at 0.3 nm. Sit at L , then compute only the rows that act there.

At this L , the formulas are:

$$\text{Re} = UL/\nu$$

$$\eta = L \text{Re}^{-3/4}$$

$$\text{Kn} = \sigma_*/L$$

η is an inertial-cascade length; only a dump if Re is large enough that $\eta < L$. Also compute η/σ_* , cohesion $(\Delta H_{\text{vap}}/N_A)/k_B T$, and, if r is near σ_* , core / attraction $(\sigma_*/r)^{12}$ vs $(\sigma_*/r)^6$. If the machine has B, ω, n , those rows too (A3, A4 in §6).

Walk the stops in this order. First one that fires is the working law.

1. Not one Newtonian $\nu > 0 \rightarrow$ Fefferman's field is not the object. Write the actual substance (^4He , plasma, ...).
2. $\text{Kn} \simeq 0.1$ or larger $\rightarrow$ continuum is dead (the pin in §5). Working law: tiles at the floor, not Navier–Stokes.
3. Cohesion unbound (boil pin, or cohesion ≈ 1 , or already a gas) $\rightarrow$ not a clinging liquid.
4. $L \approx \sigma_*$ $\rightarrow$ hydro is gone. Core vs attraction. The 12-term is the override (flat, then a wall).
5. $\text{Kn} \ll 1$ and $L \gg \sigma_*$ $\rightarrow$ hydro tile. If $\eta \geq L$, there is no inertial cascade: viscosity already won at L . If $\eta \ll L$ and $\eta \gg \sigma_*$, ordinary turbulence: the cascade dumps above the floor. If a formula gave $\eta \leq \sigma_*$, the two-term field has left the fluid; the floor has the override. At this same hydro tile, Re says inertia vs viscosity: glycerol at 1 m, 1 m/s is $\text{Re} \approx 900$ (viscosity); mercury at the same flow is $\text{Re} \approx 9 \times 10^6$ (inertia).
6. An extra gate (centrifugal, magnetic, ...) larger than the hydro row $\rightarrow$ that gate has the tile.

That is a nest event. It is not a molecule becoming infinite, and it is not Box B.

Write the verdict as one line:

material, $T \mid U, L \mid \sigma_* \mid \text{Re} \mid \eta \mid \eta/\sigma_* \mid \text{Kn} \mid \text{winner} \mid \text{NS allowed?} \mid \text{working law}$

Invert that law. If you cannot recover the pin, you have not closed it.

law	front	back
Re	$\text{Re} = UL/\nu$	$U = \text{Re } \nu/L, L = \text{Re } \nu/U$
η	$\eta = L \text{Re}^{-3/4}$	$\text{Re} = (L/\eta)^{4/3}$
Kn	$\text{Kn} = \sigma_*/L$	$L = \sigma_*/\text{Kn}$
A1	$R(\omega) = \exp(m\omega^2\Delta/(2k_B T))$	$\omega(R) = \sqrt{\ln(R) \cdot \frac{2k_B T}{m\Delta}}, R > 1$

A2, A3, A4 invert in §6. Direct check on A1: $R = 15$ returns ω , that ω returns 15 at relative error 1.3×10^{-15} .

If you need another length, change L , retile, repeat from the stops. Do not copy $(u \cdot \nabla)u$ versus $\nu \nabla^2 u$ through the floor. If the question was “smooth or blow-up for all fluids?”, stop. That question is false (Theorems 1, 2, 2'). The answer is the winner at this level.

Worked problem — water in a 1 cm pipe at 1 m/s. Pins from this paper: $\nu = 1.00 \times 10^{-6} \text{ m}^2/\text{s}$, $\sigma_* = 3.104 \times 10^{-10} \text{ m}$, $T = 293 \text{ K}$, cohesion ≈ 16 . Still a liquid. $L = 0.01 \text{ m}$, $U = 1 \text{ m/s}$.

Re	η	η/σ_*	Kn
1.00×10^4	$1.00 \times 10^{-5} \text{ m}$	3.22×10^4	3.10×10^{-8}

$\text{Kn} \ll 0.1, L \gg \sigma_*, \eta < L, \eta/\sigma_* \gg 1$. Hydro is allowed as an effective theory at 1 cm. Cascade dumps four orders above the floor. Back: $U = \text{Re } \nu/L = 1 \text{ m/s}$. Verdict: water, 20 °C | 1 m/s, 1 cm | winner = hydro (inertia/viscosity) | NS allowed at this L | floor still forbids infinite zoom.

Worked problem — air at a micron. Pins: $\nu = 1.50 \times 10^{-5} \text{ m}^2/\text{s}$, $\sigma_* = 6.5 \times 10^{-8} \text{ m}$ (mfp). $L = 1.0 \times 10^{-6} \text{ m}$, $U = 1 \text{ m/s}$.

Re	η	η/σ_*	Kn
6.67×10^{-2}	$7.62 \times 10^{-6} \text{ m}$ ($\eta > L$: no cascade)	1.17×10^2	0.065

Kn is already ≈ 0.1 . Continuum is dying. Stop at step 2. Do not evolve Navier–Stokes at this L . Working law: tiles from the mean free path. Air’s continuum is already dead at $L \approx 6.5 \times 10^{-7} \text{ m}$ (Kn = 0.1 in §5). Verdict: air, STP | 1 m/s, 1 μm | winner = Knudsen / floor | NS not allowed.

A machine with B, ω, n is the same four tools: inner and outer zones are two tiles; A1–A7 are the gates; invert as in §6. The tables in §5 are this recipe on water, air, glycerol, mercury, and ^4He at 1 m, 1 m/s. Section 7 is the nest count. python reproduce.py reprints every number in this paper, including the two worked problems above.

Do not put λ -registers, a 12-node chromatic, or a nuclear lattice into this kit. Those are other papers. This kit is floor, tile, gate, invert.

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Kolmogorov, A. N. The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers. *Dokl. Akad. Nauk SSSR* 30 (1941) 301–305. $\eta \simeq L\text{Re}^{-\frac{3}{4}}$.

Jones, J. E. On the determination of molecular fields. *Proc. R. Soc. Lond. A* 106 (1924) 463–477. Core $\simeq r^{-12}$.

CODATA 2018. k_B, N_A, e, G, m_p, m_e .

Material pins at 20 °C except air at STP: water $\nu \approx 10^{-6} \text{ m}^2/\text{s}$, $\rho = 1000 \text{ kg/m}^3$, boil 373 K; air $\nu \approx 1.5 \times 10^{-5} \text{ m}^2/\text{s}$, $\text{mfp} \approx 6.5 \times 10^{-8} \text{ m}$; glycerol $\mu \approx 1.41 \text{ Pa}\cdot\text{s}$, $\rho \approx 1260 \text{ kg/m}^3$, boil 563 K; mercury $\mu \approx 1.52 \times 10^{-3} \text{ Pa}\cdot\text{s}$, $\rho \approx 13534 \text{ kg/m}^3$, boil 630 K; ^4He λ -point 2.17 K.

Author sources. These are the author's manuscripts used as sources for this paper. Some are unpublished or private. They are still the factual citations.

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Crowell, R. D. COGNITIVE PHYSICS.

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Appendix A – map lifetime

The file `reproduce.py` reprints A1, A2, the nest work bound, the four-fluid Kolmogorov table, and the additivity check. Run `python reproduce.py`.

[illegible]

Appendix B – A1 bijection and A2 ODE

```

import math
kB, m, T = 1.380649e-23, 3.34e-27, 1.7e8
r_in, r_out, nu = 0.2, 1.0, 1e6
Delta = r_out**2 - r_in**2

def omega_from_R(R):
    return math.sqrt(math.log(R) * 2 * kB * T / (m * Delta))

def R_from_omega(w):
    return math.exp(m * w**2 * Delta / (2 * kB * T))

def tau_mig(w):
    disc = math.sqrt(nu**2 + 4*w**2)
    sp, sm = (-nu + disc)/2, (-nu - disc)/2
    A = r_in * sm / (sm - sp)
    B = r_in * sp / (sp - sm)
    lo, hi = 0.0, 1e-6
    def r(t):
        return A*math.exp(sp*t) + B*math.exp(sm*t)
    while r(hi) < r_out:
        hi *= 2
    for _ in range(80):
        mid = 0.5*(lo+hi)
        lo, hi = (mid, hi) if r(mid) < r_out else (lo, mid)
    return 0.5*(lo+hi)

# R_from_omega(omega_from_R(15)) == 15 to 1e-15
# tau_mig(omega from R(15)) == 1.340764e-6

```

Appendix C — nest additivity and work

```
import math, random
N_A, rho = 6.02214076e23, 1000.0
m = 18.01528e-3 / N_A
n = rho / m
N = n

def W(k):
    return N*(k-1)/2 + 26*N/k

# W(100)/ (N*(N-1)/2) == 1/3.36e26

random.seed(0)
k, npart = 100, 1000
vel = [[random.gauss(0,1) for _ in range(3)] for _ in range(npart)]

def PK(vs):
    P = [0.0, 0.0, 0.0]
    K = 0.0
    for v in vs:
        for a in range(3):
            P[a] += m*v[a]
        K += 0.5*m*sum(x*x for x in v)
    return P, K

P, K = PK(vel)
tiles = [PK(vel[j:j+k]) for j in range(0, npart, k)]
Psum = [sum(t[0][a] for t in tiles) for a in range(3)]
Ksum = sum(t[1] for t in tiles)
# |Psum-P|/|P| ~ 2e-15, |Ksum-K|/|K| ~ 1e-15
```